Social Media Engagement Analytics and Prediction

3–5 Minute Project Demonstration Script

0:00–0:20 — Introduction

Good evening. In this video, I will demonstrate my project, **Social Media Engagement Analytics and Prediction**.

The main objective of this project is to analyze social media engagement data, identify important patterns and engagement drivers, and provide data-driven recommendations for improving social media performance.

0:20–0:50 — Dataset

For this project, I used the **Social Media Engagement Dataset**, which contains **5,000 social media posts and 20 features**.

The dataset includes information such as:

- Platform
- Content type
- Likes
- Comments
- Shares
- Views
- Saves
- Follower count
- Engagement rate
- Posting hour
- Sentiment
- Influencer tier
- Media usage
- Verification status

The main target variable for the analysis is **Engagement_Rate**.

0:50–1:25 — Data Cleaning

The first step was data preprocessing and cleaning.

I checked the dataset for duplicate records and missing values.

I also cleaned categorical columns by removing unnecessary whitespace and converted the **Timestamp** column into the appropriate datetime format.

I verified the data types of the different columns.

Finally, I performed outlier detection using the **Interquartile Range, or IQR, method**.

Outliers were identified in variables such as likes, comments, shares, views, saves, and engagement rate. Instead of removing these records, I applied IQR-based capping to reduce the influence of extreme values while preserving the observations.

1:25–2:05 — Exploratory Data Analysis

After cleaning the data, I performed exploratory data analysis.

First, I compared the average engagement rate across different social media platforms.

Then, I analyzed different content types to understand which formats generate stronger engagement.

I also studied sentiment categories and influencer tiers to understand how these factors are associated with engagement.

Another important analysis was comparing verified and non-verified accounts, as well as posts with and without media.

Finally, I identified the top-performing posts based on engagement rate.

2:05–2:45 — Visualizations

I created several visualizations to make the analysis easier to understand.

The first visualization is a **bar chart showing average engagement rate by platform**.

The second is a **line chart showing average engagement rate by hour of the day**, which helps identify potentially stronger posting periods.

I also created a **histogram of engagement rate** to understand its distribution.

A **correlation heatmap** was used to examine relationships between numerical features.

Finally, I created a **scatter plot comparing views with engagement rate** to investigate their relationship.

2:45–3:30 — Key Findings

Based on the analysis, I found that engagement performance varies across platforms and content types.

Different content formats show different levels of audience interaction.

Engagement metrics such as likes, comments, shares, views, and saves provide useful information about post performance.

The analysis also shows that engagement varies by posting hour, which can help identify better times to publish content.

Sentiment, verification status, media usage, and influencer tier can also be compared to understand differences in engagement behavior.

I also analyzed the highest-engagement posts to identify patterns that could potentially be applied to future content strategies.

3:30–4:10 — Recommendations

Based on these findings, I recommend focusing more on platforms and content formats that demonstrate stronger average engagement.

Important content should be scheduled around the higher-performing posting hours identified from the data.

Content creators can also encourage interaction through questions, calls-to-action, and other audience engagement techniques.

Relevant hashtags and media should be used strategically.

Most importantly, social media performance should be monitored continuously using engagement rate, likes, comments, shares, views, and saves rather than relying only on follower count.

4:10–4:30 — Conclusion

To conclude, this project demonstrates how **Python, Pandas, data analysis, and visualization techniques** can be used to understand social media engagement.

The analysis provides a structured way to identify high-performing platforms, content types, posting periods, and audience interaction patterns.

Thank you for watching my project demonstration.